Approximated Riemann solver for 1D Euler equation

Project results

Fluidodinamica e ingegneria del vento computazionale

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June 2025

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Roe Scheme for 1D Euler Equations: Overview

- Defining state vector: $\mathbf{U} = \begin{bmatrix} \rho \\ \rho u \\ E \end{bmatrix}$ and flux vector: $\mathbf{F}(\mathbf{U}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ u(E+p) \end{bmatrix}$
- The conservation laws are

$$\mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0$$

- By introducing the Jacobian matrix:

$$\mathbf{A}(\mathbf{U}) = \frac{\partial \mathbf{F}}{\partial \mathbf{U}}$$

- They may be written as

$$\mathbf{U}_t + \mathbf{A}(\mathbf{U})\mathbf{U}_x = 0$$

- Replacing the Jacobian matrix $\mathbf{A}(U)$ by a constant Jacobian matrix $\tilde{\mathbf{A}} = \tilde{\mathbf{A}}(\mathbf{U}_L, \mathbf{U}_R)$

$$\mathbf{U}_t + \tilde{\mathbf{A}}(\mathbf{U}_x) = 0$$

Riemann problem vs approximate Riemann problem

Exact Riemann problem:

$$\begin{cases} \mathbf{U}_t + \mathbf{F}(\mathbf{U})_x = 0, \\ \mathbf{U}(x,0) = \begin{cases} \mathbf{U}_L & \text{if } x < 0 \\ \mathbf{U}_R & \text{if } x > 0 \end{cases} \end{cases}$$

Approximate Riemann problem:

$$\begin{cases} \mathbf{U}_t + \tilde{\mathbf{A}}\mathbf{U}_x = 0, \\ \mathbf{U}(x,0) = \begin{cases} \mathbf{U}_L & \text{if } x < 0 \\ \mathbf{U}_R & \text{if } x > 0 \end{cases} \end{cases}$$

Properties of Roe Jacobian matrix $\tilde{A}$

- Hyperbolicity of the system:
  $\tilde{A}$ is required to have real eigenvalues $\tilde{\lambda}_i = \tilde{\lambda}_i(\mathbf{U}_L, \mathbf{U}_R)$ and a complete set of linearly independent right eigenvectors.
- Consistency with the exact Jacobian:

$$\tilde{\mathbf{A}}(\mathbf{U}, \mathbf{U}) = \mathbf{A}(\mathbf{U})$$

- Conservation across discontinuities:

$$\mathbf{F}(\mathbf{U}_R) - \mathbf{F}(\mathbf{U}_L) = \tilde{\mathbf{A}}(\mathbf{U}_R - \mathbf{U}_L)$$

Roe Scheme: Averaging and Eigenstructure

- Roe scheme is a linearized Riemann solver:

$$\mathbf{F}_{i+\frac{1}{2}} = \frac{1}{2}(\mathbf{F}_L + \mathbf{F}_R) - \frac{1}{2}\sum_{i=1}^{m} \tilde{\alpha}_i |\tilde{\lambda}_i| \tilde{\mathbf{K}}^{(i)}.$$

- Roe averages between left (L) and right (R) states:

$$\tilde{u} = \frac{\sqrt{\rho_L} u_L + \sqrt{\rho_R} u_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \quad \tilde{H} = \frac{\sqrt{\rho_L} H_L + \sqrt{\rho_R} H_R}{\sqrt{\rho_L} + \sqrt{\rho_R}}, \quad \tilde{a} = \sqrt{(\gamma - 1)\left(\tilde{H} - \frac{1}{2}\tilde{u}^2\right)}$$

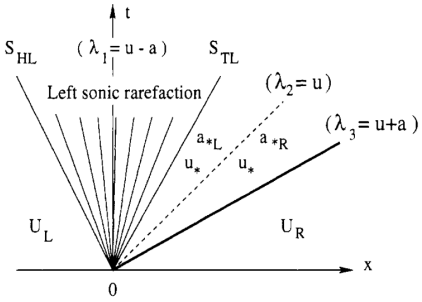
- Approximate $\mathbf{A}$ and compute:
  - Eigenvectors $\mathbf{K}$
  - Eigenvalues $\{\tilde{u} - \tilde{a}, \tilde{u}, \tilde{u} + \tilde{a}\}$
  - wave strengths $\tilde{\alpha}_i$ solution of $\Delta \mathbf{U} = \sum_{i=1}^{3} \tilde{\alpha}_i \tilde{\mathbf{K}}^{(i)}$

Harten-Hyman Entropy Fix: Motivation

- Roe's linearized solver may violate entropy conditions in transonic rarefactions.
- In the case of a left transonic rarefaction (specular for right case):

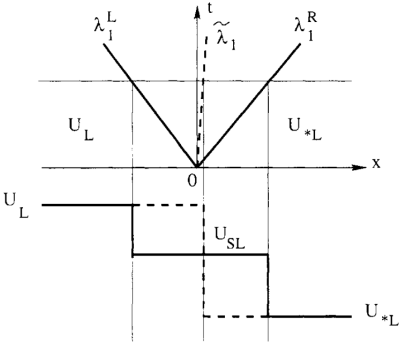
$$\lambda_1(U_L) = S_{HL} = u_L - a_L < 0$$

$$\lambda_1(U_{*L}) = S_{TL} = u_* - a_{*L} > 0$$



Entropy Fix: Modified Roe Flux

- Big jump travelling with speed $\tilde{\lambda}_1$ is split into two smaller jumps of speed λ_1^L and λ_1^R.
- From conservation laws we find U_{SL}.
- We use the same flux formula of Roe applied to $U_{SL} - U_L$.
- Actually, doing math we can just define a new $\bar{\lambda}_1$ and use Roe formula for fluxes.



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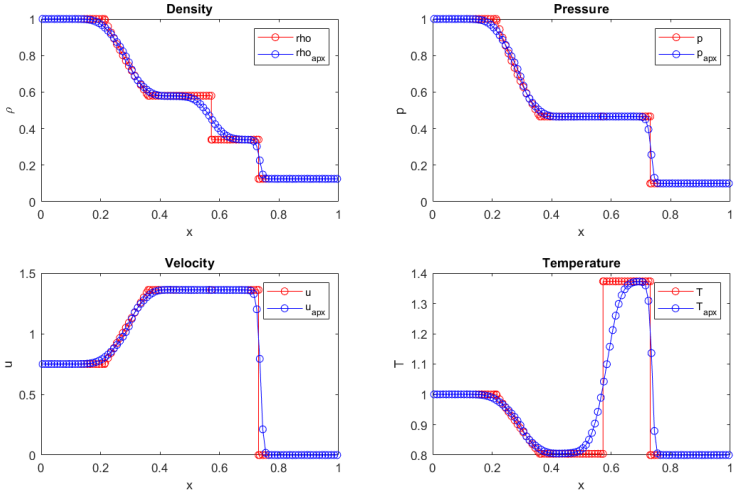
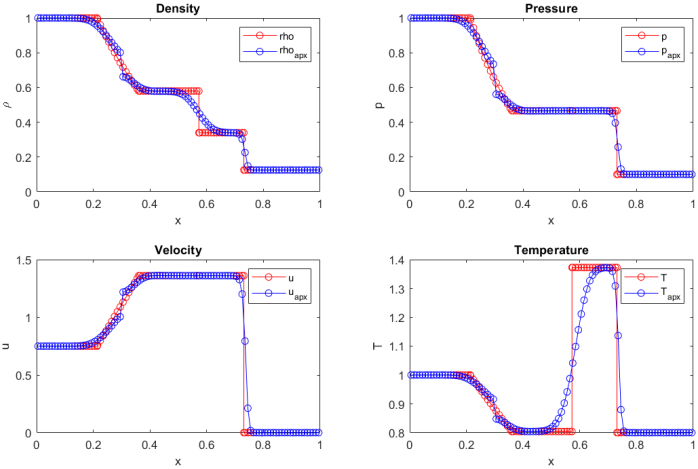
Test 1

2 Test Result



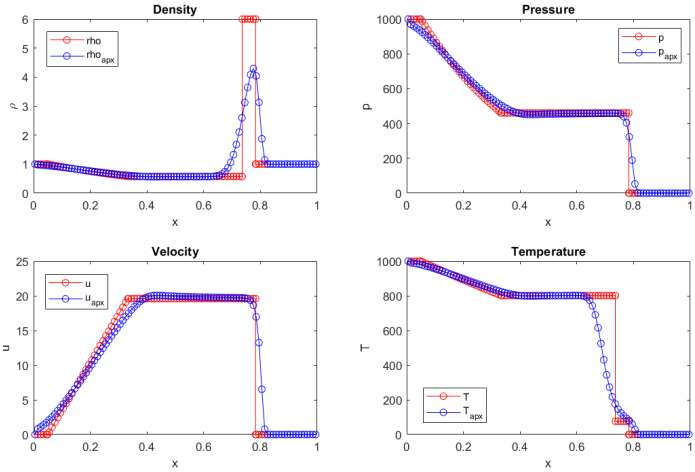
Test 1 with no fixing point

2 Test Result



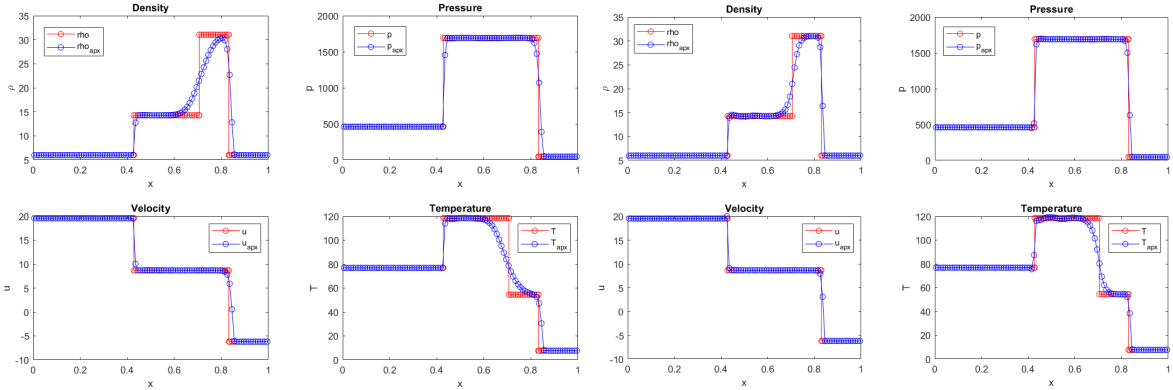
Test 3

2 Test Result



Test 4

2 Test Result

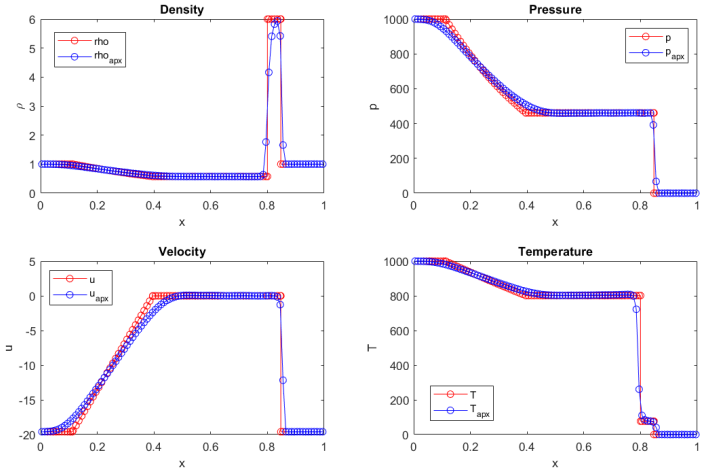


Order 1

Order 2

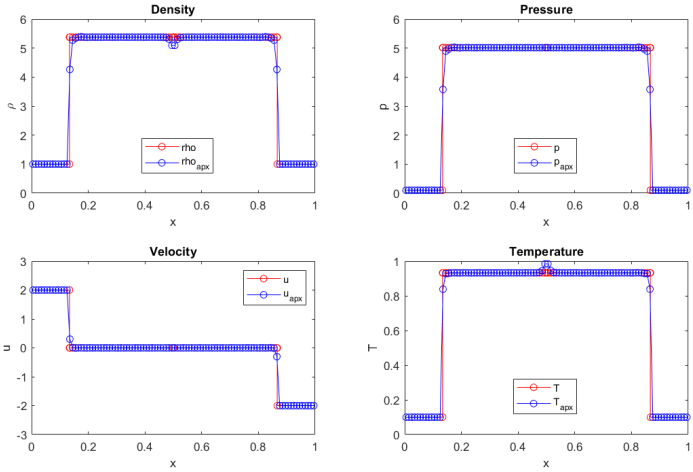
Test 5

2 Test Result



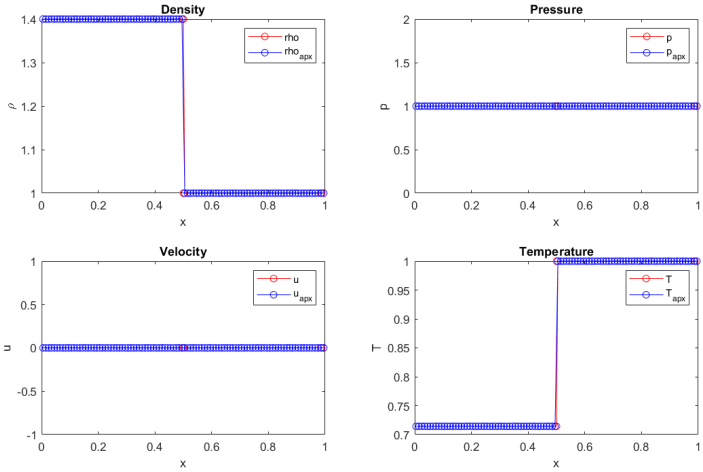
Test 6

2 Test Result



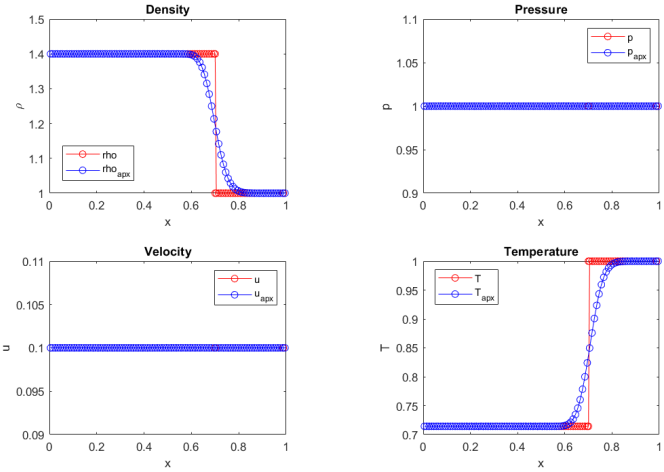
Test 7

2 Test Result



Test 8

2 Test Result



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Final Observations

- No need of an exact Riemann solver.
- Computational efficiency of approximated Riemann solver.
- Well behavior in the worst case scenario as the proposed test.

Approximated Riemann solver for 1D Euler equation

Thank you for listening!
Any questions?

Backup Slides

Diagonalisation and Characteristic Variables

- Diagonalisable System:

$$\mathbf{A} = \mathbf{K}\Lambda\mathbf{K}^{-1} \quad \textit{or} \quad \Lambda = \mathbf{K}^{-1}\mathbf{A}\mathbf{K}$$

- Characteristic variables:

$$\mathbf{W} = \mathbf{K}^{-1}\mathbf{U} \quad \textit{or} \quad \mathbf{U} = \mathbf{K}\mathbf{W}$$

- Canonical form or characteristic form of hyperbolic PDEs system:

$$\mathbf{W}_t + \Lambda\mathbf{W}_x = 0$$

$$\frac{\partial \omega_i}{\partial t} + \lambda_i \frac{\partial \omega_i}{\partial x} = 0, \quad i = 1, \ldots, m$$

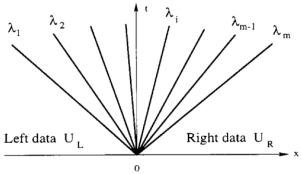
Riemann problem's solution

- The solution of the general Initial Value Problem:

$$\mathbf{U}(x,t) = \sum_{i=1}^{m} \omega_i(x,t)\mathbf{K}^{(i)}$$

- Visualization of the Riemann problem:

$$\mathbf{U_L} = \sum_{i=1}^{m} \alpha_i \mathbf{K}^{(i)}, \quad \mathbf{U_R} = \sum_{i=1}^{m} \beta_i \mathbf{K}^{(i)}$$

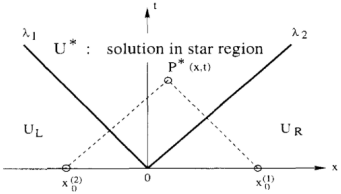


- Final solution to the Riemann problem in terms of the original variables:

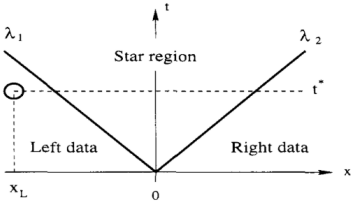
$$\mathbf{U}(x,t) = \sum_{i=I+1}^{m} \alpha_i \mathbf{K}^{(i)} + \sum_{i=1}^{I} \beta_i \mathbf{K}^{(i)}$$

Solution for a 2 x 2 System

4 Backup Slides



Structure of the solution of the Riemann problem for a 2 x 2 linear system with constant coefficients



Riemann problem solution found by travelling along dashed horizontal line t = t*

- Solution in the Star Region, between the λ_1 and λ_2

$$\mathbf{U}^*(x,t) = \beta_1 \mathbf{K}^{(1)} + \alpha_2 \mathbf{K}^{(2)}$$

- Jump in U across wave i

$$(\Delta \mathbf{U})_i = (\beta_i - \alpha_i)\mathbf{K}^{(i)}$$

References